Unit - I : Set Theory & Mathematical logic

Defⁿ: A set is a well define collection or class of distinct objects / elements / members.

By a well-define collection we means that there exist a rule with the help of which we should be able to say whether any given elements belongs to or not belongs to the collection under consideration.

For eg. ① set of even positive integers less than 12
i.e. 2, 4, 6, 8, 10

② set of students in a class with first name starting with alphabet S.

Generally capital letters such as A, B, C, X, Y ..etc. are used to denote the sets. and their members are denoted by small case letters such as a, b, ..x, y etc.

Representation of sets :

① Roster Method / List method

In this method elements of a set are described by listing or writing them exahoustively.

for eg: ① set of vowels in the English alphabet.

$$V = \{a, e, i, o, u\}$$

(ii) set of Natural number not greater than 10

$$N = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

Scanned by CamScanner

(ii) Set Selection method :

In this method the elements of a set are represented by stating their properties (qualitative, quantitative or both) which uniquely characterize them.

For ex: (1) set of Natual number is denoted as

$$N = \{ x \mid x > 0, \& \ x \in I \ (\text{Integers}) \}$$

(2) Set of all even ~~integers~~ (positive) integers

$$I_+ = \{ 2, 4, 6, 8 \ldots \}$$

$$= \{ x \mid x \in I, \ x \text{ is divisible by } 2 \}$$

Types of sets :

(1) Finite and Infinite set :

A set is said to be a finite set if it contain finite number of distinct elements. otherwise it is known as infinite set

For eg: (1) set of ~~all~~ english alphabets. is a finite set

(ii) set of Natural number is ~~an~~ Infinite set

(2) Singleton Set :

A set which contain only one element is called a singleton (or unit) set.

For eg:: (1) $A = \{ x \mid 4 < x < 6, \ x \text{ is integer} \}$

$$= \{ 5 \}$$

(2) $I = \{ x \mid x^2 = 9 \ \& \ x < 0 \}$

$$= \{ -3 \}$$

Scanned by CamScanner

(iii) Null set / Empty set / void set :

A set which do not contain any element is called a null set / Empty set / void set.

This set is denoted by Greek letter ϕ (phi)

For eg: (1) ϕ = set of all x which is not equal to x itself

$= \{ x / x \neq x \}$

$\therefore \phi = \{ \}$

(ii) set of all integers whose square is 5

$\phi = \{ x / x^2 = 5, x \in I \}$

$\phi = \{ \}$ since there is no such +ve integer whose square is 5

Equality of sets :

Two sets A & B are said to be equal if all elements of set A are same as elements of B also having same number of elements in each sets.

Symbolically two equal sets denoted as

$$A = B \Leftrightarrow (A \subseteq B \ \& \ B \subseteq A)$$

ie $A = B$ if and only if $x \in A \Leftrightarrow x \in B$.

This is known as Axiom of extension or identity

For eg: (1) If $A = \{4, 3, 2, 1\}$ & $B = \{1, 3, 2, 4\}$

Then $A = B$.

(2) If $A = \{2, 3\}$, $B = \{3, 2\}$ & $C = \{x / x^2 - 5x + 6 = 0\}$

Then $A = B = C$

Scanned by CamScanner

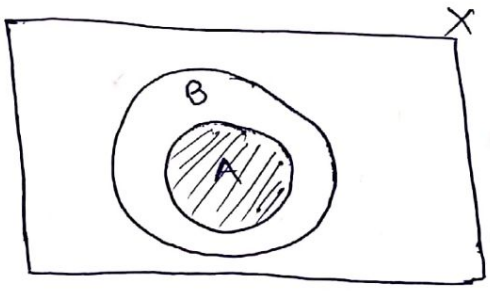
★ Def^n : Subset :

Let A & B are two non empty sets.
Then set A is said to be a subset of set B iff every element of set A is an element of set B.
and it is denoted as $\boxed{A \subseteq B}$

Here we ~~also~~ can also read it as
" A is contained in B"
OR "B contains A"

ie. $\forall x$, $x \in A \implies x \in B$
$\iff A \subseteq B$.

Venn Diagram



Here B is superset of ~~A~~ set A (ie. B contains A)

Note: If atleast one element of set A is not ~~can~~ belong to the set B. then set A will not be a subset of B. ie. $A \not\subseteq B$

ie. $x \in A \implies x \notin B$

Note: (1) An empty set ϕ is always a subset of every set. ie. $\phi \subseteq A$

(2) Every set is a subset of itself
ie $A \subseteq A$

Scanned by CamScanner

Defn: Proper Subset :

A set A is said to be a proper subset of set B. (ie A is properly contain in set B) iff it satisfy followings.

① Every element of set A is also an element of set B
ie. $A \subseteq B$

② There exist atleas one element in set B such that this element is not belongs to set A.
ie. for $y \in B$ s.t. $y \notin A$.
i.e $A \neq B$.

Then A is proper subset of B. and symbolically it is denoted as $\boxed{A \subset B}$

★ **Family of Sets &(Power set)**

"If ~~the~~ elements of the set are sets themselves then such set is known as family of sets."

↳ If A is any set then set of all subsets of set A is known as power set of A. denoted as P(A)

For eg: let $A = \{a, b, c\}$

Then $P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$

★ **Universal set :** In application of set theory all the sets under discussion are assumed to be the subset of a fixed set. This fixed set is called as universal set OR univers of discourse.

This set is usually denoted by U OR X

Scanned by CamScanner

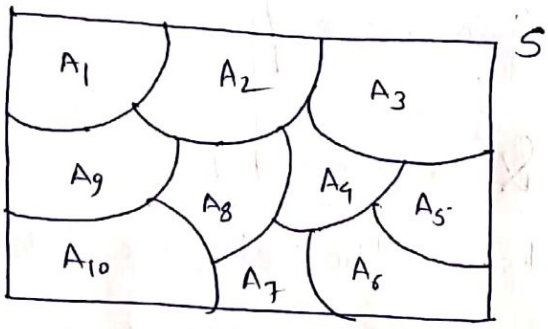
✱ Partitions :

Let S be a non-empty set, A partition of S is a subdivision of set S into non-overlaping (disjoint) non-empty subsets.

Precisely, a partition of S is collection of all non empty {Ai} ~~subset~~ disjoint subsets of S.

Such that
- (i) each $a \in S$, belongs to one of subset A_i of S only.
- (ii) let A_i & A_j are two subsets of S Then they must be disjoint

  ie. $A_i \cap A_j = \phi$, $i \neq j$

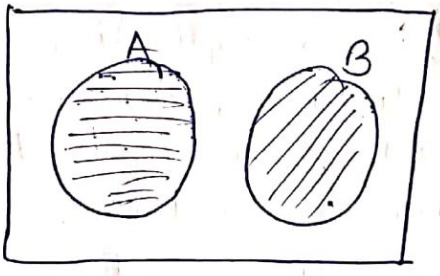


✱ Disjoint set (Mutually Exclusive)

let A & B are any two non-empty sets, these two sets are said to be disjoint sets or mutually exclusive if both sets A & B does not have any common element.

ie $A \cap B = \phi$.

Venn Diagram :



Scanned by CamScanner

Basic Operations of Set :

In set theory, union, intersection, complement, difference are the basic operation, which are defined as follows.

① Complement of set

Let A be any set, then the complement of set A is defined and denoted as

$$\overline{A} = A^c = U - A$$

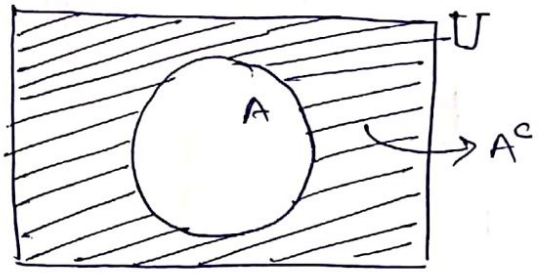
where, U is universal set.

i.e. Complement of set A is the set of all element of Universal set which are not belongs to set A.

ie. let $x \in U$ & $x \notin A$

$\Rightarrow x \in A^c$

Venn Diagram :



For eg : let $U = \{a, b, c, d, e, f, g, h\}$ be the universal set

and $A = \{a, c, e, g\}$ be the subset of U

Then $A^c = \{b, d, f, h\}$ is the complement of A.

Note :

(i) $U^c = \phi$

(ii) $\phi^c = U$

(iii) $(A^c)^c = A$

(iv) $A \cup A^c = U$

(v) $A \cap A^c = \phi$

Scanned by CamScanner

(11) Union of sets

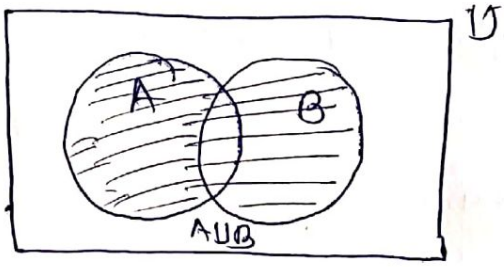
Let A & B are any two non-empty sets then their union is the set of all elements which either belongs to set A OR belongs to set B OR belongs to both A & B.

Symbolically it is denoted and define as

$$A \cup B = \{ x \mid x \in A \text{ OR } x \in B \text{ OR } x \in A \& B \text{ both} \}$$

Here a word 'OR' is used in the sense of 'and / or'

Venn Diagram :



Some important Rules

(i) $A \cup A = A$

(ii) $A \cup \phi = A$ ($\because \phi$ is empty set)

(iii) $A \cup U = U$, ($\because U$ is universal set ie. $A \subseteq U$)

(iv) $A \subseteq A \cup B$ & $B \subseteq A \cup B$

(v) If $A \subseteq B$ then $A \cup B = B$

Scanned by CamScanner

(iii) **Intersection of sets :**

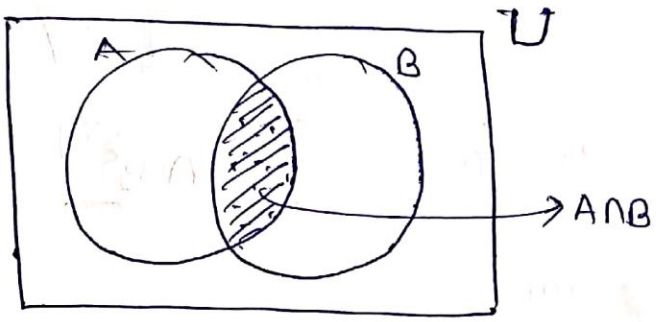
let A & B are any two non-empty sets then their intersection is the set of all those element which belong to ~~both~~ set A & B. both.

(ie. Intersection of A & B is the set of common elements in A & B.)

Symbolically it is denoted and define as

$$A \cap B = \{ x / x \in A \text{ and } x \in B \}$$

Venn Diagram :



Some important Rule

(i) If $A \subseteq B$ Then $A \cap B = A$

(ii) Since $A \subseteq U$, $A \cap U = A$

(iii) Since $\phi \subseteq A$, $\therefore A \cap \phi = \phi$

(iv) $A \cap B \subseteq A \cup B$

(v) $A \cap B \subseteq A$

$\subseteq B$

Scanned by CamScanner

★ Difference of two sets :

If A and B are any two sets, then the difference of A and B is the set containing elements which belongs to set A but not belongs to set B.
Symbolically it is denoted and define as

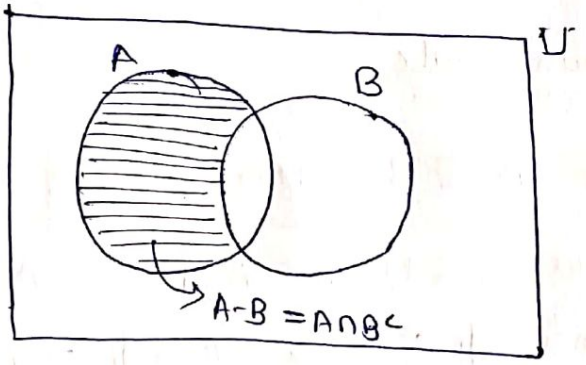
$$A - B = A \sim B = \{x / x \in A \ \& \ x \notin B\}$$

$$= \{x / x \in A \ \& \ x \in B^c\}$$

$$= \{x / x \in (A \cap B^c)\}$$

$$\Rightarrow \boxed{A - B = A \cap B^c}$$

Venn Diagram :



Some important Rules :

(1) $A - A = \phi$

(2) $A - \phi = A$

(3) $A - B \subseteq A$

Scanned by CamScanner

Example ① The union of sets is commutative.

If A and B are any two sets then $A \cup B = B \cup A$

Soln: let A & B are any two non-empty sets

To show $A \cup B = B \cup A$

we need to prove that (i) $A \cup B \subseteq B \cup A$
& (ii) $B \cup A \subseteq A \cup B$

To prove (i) $A \cup B \subseteq B \cup A$

let x be the any arbitrary element such that

$x \in A \cup B \Rightarrow x \in A$ OR $x \in B$

$\Rightarrow x \in B$ OR $x \in A$

$\therefore$ for $x \in A \cup B \Rightarrow x \in B \cup A$

$\Rightarrow \boxed{A \cup B \subseteq B \cup A}$ ——— (1)

To prove (ii) $B \cup A \subseteq A \cup B$

let y be the any arbitrary element such that

$y \in B \cup A \Rightarrow y \in B$ OR $y \in A$

$\Rightarrow y \in A$ OR $y \in B$

$\Rightarrow y \in A \cup B$

$\therefore$ For $y \in B \cup A \Rightarrow y \in A \cup B$

$\therefore \boxed{B \cup A \subseteq A \cup B}$ ——— (2)

From ① & ②:

Since $A \cup B \subseteq B \cup A$
& $B \cup A \subseteq A \cup B$

$\Rightarrow \boxed{A \cup B = B \cup A}$

i.e. The union of sets is commutative

Scanned by CamScanner

Example (2): Intersection of sets is commutative

i.e. If A & B are any two nonempty sets

Then $A \cap B = B \cap A$.

Soln: let A & B are any two non-empty set

To show $A \cap B = B \cap A$

we need to prove that (i) $A \cap B \subseteq B \cap A$

& (ii) $B \cap A \subseteq A \cap B$

To prove (i) $A \cap B \subseteq B \cap A$

let x be the any arbitrary element such that

$x \in A \cap B \Rightarrow x \in A$ and $x \in B$

$\Rightarrow x \in B$ and $x \in A$

$\Rightarrow x \in (B \cap A)$

Since for $x \in A \cap B \Rightarrow x \in B \cap A$

$\therefore \boxed{A \cap B \subseteq B \cap A}$ ——— (1)

To prove (ii) $B \cap A \subseteq A \cap B$

let y be the any arbitrary element such that

$y \in B \cap A \Rightarrow y \in B$ and $y \in A$

$\Rightarrow y \in A$ and $y \in B$

$\Rightarrow y \in (A \cap B)$

$\therefore$ Since for $y \in (B \cap A) \Rightarrow y \in (A \cap B)$

$\therefore \boxed{(B \cap A) \subseteq (A \cap B)}$ ——— (2)

$\therefore$ from (1) & (2)

$\boxed{A \cap B = B \cap A}$ H.P.

$\therefore$ Intersection of sets is commutative.

Scanned by CamScanner

Example ③: Union of sets is Associative

If A, B & C are any three non-empty sets then

$$A \cup (B \cup C) = (A \cup B) \cup C$$

Soln: Let x be the any arbitrary element such that

For $x \in A \cup (B \cup C) \Rightarrow x \in A$ OR $x \in (B \cup C)$

$\Rightarrow x \in A$ OR $(x \in B$ OR $x \in C)$

$\Rightarrow (x \in A$ OR $x \in B)$ OR $x \in C$

$\Rightarrow x \in (A \cup B)$ OR $x \in C$

$\Rightarrow x \in (A \cup B) \cup C$

$\therefore \boxed{A \cup (B \cup C) \subseteq (A \cup B) \cup C}$ ——— ①

Now, let us consider y be the any arbitrary element such that

$y \in (A \cup B) \cup C \Rightarrow y \in (A \cup B)$ OR $y \in C$

$\Rightarrow (y \in A$ OR $y \in B)$ OR $y \in C$

$\Rightarrow y \in A$ OR $(y \in B$ OR $y \in C)$

$\Rightarrow y \in A$ OR $y \in (B \cup C)$

$\Rightarrow y \in A \cup (B \cup C)$

$\therefore \boxed{(A \cup B) \cup C \subseteq A \cup (B \cup C)}$ ——— ②

From ① & ②, since

$A \cup (B \cup C) \subseteq (A \cup B) \cup C$

& $(A \cup B) \cup C \subseteq A \cup (B \cup C)$

$\therefore \boxed{A \cup (B \cup C) = (A \cup B) \cup C}$ H.P

Hence Union of sets is Associative.

Scanned by CamScanner

Example (4) The intersections of sets is associative

let A, B & C are any three non-empty sets then $A \cap (B \cap C) = (A \cap B) \cap C$

Soln: let A, B & C are any three non-empty sets. and let x be the any arbitrary element such that

$x \in A \cap (B \cap C) \Rightarrow x \in A$ and $x \in (B \cap C)$

$\Rightarrow x \in A$ and $(x \in B$ and $x \in C)$

$\Rightarrow (x \in A$ and $x \in B)$ and $x \in C$

$\Rightarrow x \in A \cap B$ and $x \in C$

$\Rightarrow x \in (A \cap B) \cap C$

$\therefore \boxed{A \cap (B \cap C) \subseteq (A \cap B) \cap C}$ ——— (1)

Similarly let y be the any arbitrary element such that

$y \in (A \cap B) \cap C \Rightarrow y \in A \cap B$ and $y \in C$

$\Rightarrow (y \in A$ and $y \in B)$ and $(y \in C)$

$\Rightarrow y \in A$ and $(y \in B$ and $y \in C)$

$\Rightarrow y \in A$ and $y \in (B \cap C)$

$\Rightarrow y \in A \cap (B \cap C)$

$\therefore \boxed{(A \cap B) \cap C = A \cap (B \cap C)}$ ——— (2)

from (1) & (2) Since

$(A \cap B) \cap C \subseteq A \cap (B \cap C)$

& $A \cap (B \cap C) \subseteq (A \cap B) \cap C$

$\therefore \boxed{A \cap (B \cap C) = (A \cap B) \cap C}$ H.P.

Scanned by CamScanner

Example (5): prove that

(i) $A - B = A \cap B^c$

(ii) If $A \subseteq B$ then $A \cup (B - A) = B$

Soln: To prove (i) $A - B = A \cap B^c$

Let x be the any arbitrary element such that

$x \in A - B \iff x \in A$ and $x \notin B$

$\iff x \in A$ and $x \in B^c$

$\iff x \in (A \cap B^c)$

$\therefore \boxed{A - B = A \cap B^c}$ H.P.

Now. To prove (ii) If $A \subseteq B$ then $A \cup (B - A) = B$

Let $A \subseteq B$ i.e. for any element

$x \in A \Rightarrow x \in B$ ——(1)

Now. consider

$x \in A \cup (B - A)$

$\iff x \in A$ OR $x \in (B - A)$

$\iff x \in A$ OR $(x \in B$ and $x \notin A)$

$\iff x \in A$ OR $(x \in B$ and $x \in A^c)$

$\iff (x \in A$ OR $x \in B)$ and $(x \in A$ OR $x \in A^c)$

$\iff x \in A \cup B$ and $x \in A \cup A^c$

$\iff x \in A \cup B$ and $x \in U$ $\because (A \cup A^c = U)$

$\iff x \in [(A \cup B) \cap U]$

$\iff x \in (A \cup B)$

$\iff x \in B$ $\because \boxed{A \subseteq B \Rightarrow A \cup B = B}$

$\therefore \boxed{A \cup (B - A) = B}$ H.P.

Scanned by CamScanner

Example ⑥: ~~Show that~~, for any Non-empty sets A, ~~&~~ ~~c~~ B show that

(a) $A - B$, $A \cap B$ & $B - A$ are mutually disjoint

(b) $(A - B) \cap B = \phi$

(c) $(A - B) \cup A = A$

Soln: (a) To Prove: $A - B$, $A \cap B$ & $B - A$ are mutually disjoint.

Let A & B are any two non empty sets and x be the any arbitrary element so that

$x \in (A - B) \cap (A \cap B)$

$\Leftrightarrow x \in (A - B)$ and $x \in (A \cap B)$

$\Leftrightarrow (x \in A$ and $x \notin B)$ and $(x \in A$ and $x \in B)$

$\Leftrightarrow x \in A$ and $x \in B$ and $x \notin B$

$\Leftrightarrow x \in A$ and $(x \in B$ and $x \in B^c)$

$\Leftrightarrow x \in A$ and $x \in (B \cap B^c)$

$\Leftrightarrow x \in A$ and $x \in \phi$

$\Leftrightarrow x \in A \cap \phi$

$\Leftrightarrow x \in \phi$.

$\therefore \boxed{(A - B) \cap (A \cap B) = \phi}$ ——— (1)

i.e. $A - B$ & $A \cap B$ are mutually disjoint sets.

Similarly.

let y be the any arbitrary element such that

$y \in (A \cap B) \cap (B - A)$

$\Leftrightarrow y \in (A \cap B)$ and $y \in (B - A)$

$\Leftrightarrow (y \in A$ and $y \in B)$ and $(y \in B$ and $y \notin A)$

Scanned by CamScanner

$\Leftrightarrow y \in A$ and $y \in B$ and $y \notin B$

$\Leftrightarrow y \in A$ and $y \in B$ and $y \in B^c$

$\Leftrightarrow y \in A$ and $y \in (B \cap B^c)$

$\Leftrightarrow y \in A$ and $y \in \phi$

$\Leftrightarrow y \in (A \cap \phi)$

$\Leftrightarrow y \in \phi$

$\therefore \boxed{(A \cap B) \cap (B - A) = \phi}$ ——— (2)

This shows that $A \cap B$ & $B - A$ are mutually disjoint sets.

also, consider z be the any arbitrary element such that

$z \in (A - B) \cap (B - A)$

$\Leftrightarrow z \in (A - B)$ and $z \in (B - A)$

$\Leftrightarrow (z \in A$ and $z \notin B)$ and $(z \in B$ and $z \notin A)$

$\Leftrightarrow (z \in A$ and $z \notin A)$ and $(z \in B$ and $z \notin B)$

$\Leftrightarrow (z \in A$ and $z \in A^c)$ and $(z \in B$ and $z \in B^c)$

$\Leftrightarrow z \in (A \cap A^c)$ and $z \in (B \cap B^c)$

$\Leftrightarrow z \in \phi$ and $z \in \phi$

$\Leftrightarrow z \in \phi$

$\therefore \boxed{(A - B) \cap (B - A) = \phi}$ ——— (3)

This shows that $(A - B)$ & $(B - A)$ are mutually disjoint sets

$\therefore$ from eqn (1), (2) & (3),

$A - B$, $A \cap B$ & $B - A$ are mutually disjoint sets.

H.P.

Scanned by CamScanner

(b) To prove: $(A-B) \cap B = \phi$

Let x be the any arbitrary element such that

$x \in (A-B) \cap B$

$\Leftrightarrow x \in A-B$ and $x \in B$

$\Leftrightarrow (x \in A$ and $x \notin B)$ and $(x \in B)$

$\Leftrightarrow x \in A$ and $(x \notin B$ and $x \in B)$

$\Leftrightarrow x \in A$ and $(x \in B^c$ and $x \in B)$

$\Leftrightarrow x \in A$ and $x \in (B^c \cap B)$

$\Leftrightarrow x \in A$ and $x \in \phi$

$\Leftrightarrow x \in (A \cap \phi)$

$\Leftrightarrow x \in \phi$

$\therefore \boxed{(A-B) \cap B = \phi}$ H.P.

(c) To show $(A-B) \cup A = A$

Let x be the any arbitrary element such that

$x \in (A-B) \cup A$

$\Leftrightarrow x \in (A-B)$ OR $x \in A$

$\Leftrightarrow (x \in A$ and $x \notin B)$ OR $x \in A$

$\Leftrightarrow x \in A$ and $(x \notin B$ OR $x \in A)$

$\Leftrightarrow x \in A$ and $(x \in B^c$ OR $x \in A)$

$\Leftrightarrow x \in A$ and $x \in B^c$

This shows that x is always an element of A

ie $\Leftrightarrow x \in A$

$\therefore \boxed{(A-B) \cup A = A}$ H.P.

Scanned by CamScanner

Example (7) (Distributive law's)

If A, B & C are any three non-empty sets. then

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ } [Left Distributive law]

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ }

& (c) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ } (Right Distributive law)

(d) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$ }

Proof: To prove: (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

let x be the any arbitrary element such that

$x \in A \cup (B \cap C)$

$\Leftrightarrow (x \in A)$ OR $x \in (B \cap C)$

$\Leftrightarrow (x \in A)$ OR $(x \in B$ and $x \in C)$

$\Leftrightarrow (x \in A$ OR $x \in B)$ and $(x \in A$ OR $x \in C)$

$\Leftrightarrow (x \in A \cup B)$ and $(x \in A \cup C)$

$\Leftrightarrow x \in (A \cup B) \cap (A \cup C)$

$\therefore$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ H.P.

To prove (c) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

let x be the any arbitrary element such that

$x \in (A \cup B) \cap C$

$\Leftrightarrow (x \in A \cup B)$ and $(x \in C)$

$\Leftrightarrow (x \in A$ OR $x \in B)$ and $(x \in C)$

$\Leftrightarrow (x \in A$ and $x \in C)$ OR $(x \in B$ and $x \in C)$

$\Leftrightarrow (x \in A \cap C)$ OR $(x \in B \cap C)$

$\Leftrightarrow x \in (A \cap C) \cup (B \cap C)$

Scanned by CamScanner

$\therefore$ $\boxed{(A \cup B) \cap C = (A \cap C) \cup (B \cap C)}$ H.P.

Note: proof of (b) & (d) left for the readers

Example (3) (De-Morgan's law)

(a) Show that

(a) $(A \cup B)^c = A^c \cap B^c$

(b) $(A \cap B)^c = A^c \cup B^c$.

Soln: To prove (a) $(A \cup B)^c = A^c \cap B^c$:

Let x be the any arbitrary element such that

$x \in (A \cup B)^c \iff x \notin (A \cup B)$

$\iff x \notin A$ and $x \notin B$

$\iff x \in A^c$ and $x \in B^c$

$\iff x \in (A^c \cap B^c)$

$\therefore$ $\boxed{(A \cup B)^c = A^c \cap B^c}$ H.P.

Now. To prove $(A \cap B)^c = A^c \cup B^c$

Let x be the any arbitrary element such that

$x \in (A \cap B)^c \iff x \notin (A \cap B)$

$\iff x \notin A$ OR $x \notin B$

$\iff x \in A^c$ OR $x \in B^c$

$\iff x \in (A^c \cup B^c)$

$\therefore$ $\boxed{(A \cap B)^c = A^c \cup B^c}$ H.P.

Scanned by CamScanner

Ex (9) Let A, B & C are any three non-empty sets. Then show that

(a) $A-(B\cup C) = (A-B)\cap(A-C)$

(b) $A-(B\cap C) = (A-B)\cup(A-C)$

Soln: (a) To prove $A-(B\cup C) = (A-B)\cap(A-C)$

Let x be the any arbitrary element such that

$x \in A-(B\cup C) \iff x\in A$ and $x\notin (B\cup C)$

$\iff x\in A$ and $x\in (B\cup C)^c$

$\iff x\in A$ and $x\in (B^c\cap C^c)$ (∵ By De-Morgan law)

$\iff x\in A$ and $(x\in B^c$ and $x\in C^c)$

$\iff (x\in A$ and $x\in B^c)$ and $(x\in A$ and $x\in C^c)$

$\iff (x\in A$ and $x\notin B)$ and $(x\in A$ and $x\notin C)$

$\iff [x\in A-B]$ and $[x\in A-C]$

$\iff x\in (A-B)\cap(A-C)$

$\therefore \boxed{A-(B\cup C) = (A-B)\cap(A-C)}$ H.P.

(b) To prove $A-(B\cap C) = (A-B)\cup(A-C)$

Let x be the any arbitrary element such that

$x \in A-(B\cap C) \iff x\in A$ and $x\notin (B\cap C)$

$\iff x\in A$ and $x\in (B\cap C)^c$

$\iff x\in A$ and $x\in (B^c\cup C^c)$ (∵ By Demorgan law)

$\iff (x\in A$ and $x\in B^c)$ or $(x\in A$ or $x\in C^c)$

$\iff (x\in A$ and $x\notin B)$ or $(x\in A$ and $x\notin C)$

$\iff (x\in A-B)$ or $(x\in A-C)$

$\iff x\in (A-B)\cup(A-C)$

$\therefore \boxed{A-(B\cap C) = (A-B)\cup(A-C)}$ H.P.

Scanned by CamScanner

Example (10) If A and B are any two sets, then show that

(a) $A - B = A \iff A \cap B = \phi$

(b) $A - B = \phi \iff A \subseteq B$

(c) $A \cup B = A \cap B \iff A = B$.

Soln: (a) **To show** $A - B = A \iff A \cap B = \phi$

Let us consider $A - B = A$ ——— (1)

we now show that $A \cap B = \phi$

Let x be any arbitrary element such that

$x \in A \cap B \iff x \in A$ and $x \in B$

$\iff x \in (A - B)$ and $x \in B$ (∵ from (1))

$\iff (x \in A$ and $x \notin B)$ and $x \in B$

$\iff x \in A$ and $(x \in B^c$ and $x \in B)$

$\iff (x \in A)$ and $(x \in B^c \cap B)$

$\iff x \in A$ and $x \in \phi$

$\iff x \in A \cap \phi$

$\iff x \in \phi$

$\therefore \boxed{A \cap B = \phi}$ H.P.

Conversely: Let us consider $A \cap B = \phi$, ——— (2)

we show that $A - B = A$

Let x be any arbitrary element such that

$x \in A - B \iff x \in A$ and $x \notin B$

$\iff x \in A$ and $x \in B^c$

$\iff x \in (A \cap B^c)$

$\iff x \in (A \cap B^c) \cap \phi$ ∵ for any nonempty set X, $X \cap \phi = X$

$\iff x \in (A \cap B^c)$ and $x \in \phi$

Scanned by CamScanner

$\Leftrightarrow [x \in A \cap B^c]$ and $[x \in A \cap B]$ (∵ By ②)

$\Leftrightarrow [x \in A$ and $x \in B^c]$ and $[x \in A$ and $x \in B]$

$\Leftrightarrow x \in A$ and $x \in B^c$ and $x \in B$

$\Leftrightarrow x \in A$ and $x \in B^c \cap B$

$\Leftrightarrow x \in A$ and $x \in \phi$

$\Leftrightarrow x \in A \cap \phi$

$\Leftrightarrow x \in A$

$\therefore \boxed{A - B = A}$ Hip.

(b) To Show $A - B = \phi \Leftrightarrow A \subseteq B$

Let us consider that $A - B = \phi$ ——— ③

To we show that $A \subseteq B$

let X be the universal set, and A & B both are subset of X. (ie. $A \subseteq X$ & $B \subseteq X$)

let x be the any arbitrary element in X. such that

$x \in A \Rightarrow x \in A \cap X$ (∵ since $A \subseteq X$, $A \cap X = A$)

$\Rightarrow x \in A$ and $x \in X$

$\Rightarrow x \in A$ and $x \in (B \cup B^c)$ (∵ since $B \subseteq X$, $B \cup B^c = X$)

$\Rightarrow x \in A$ and $(x \in B$ OR $x \in B^c)$

$\Rightarrow [x \in A$ and $x \in B]$ OR $[x \in A$ and $x \in B^c]$

$\Rightarrow [x \in A$ and $x \in B]$ OR $[x \in A$ and $x \notin B]$

$\Rightarrow (x \in A \cap B)$ OR $(x \in A - B)$

$\Rightarrow x \in A \cap B$ OR $x \in \phi$ (∵ given ③)

$\Rightarrow x \in (A \cap B) \cup \phi$

$\Rightarrow x \in A \cap B$

$\Rightarrow x \in A$ and $x \in B$

$\Rightarrow x \in B$ $\therefore \boxed{A \subseteq B}$ Hip.

Scanned by CamScanner

Conversely, consider $A \subseteq B$ ——(4)

we show that $A - B = \phi$

let $x \in A - B \iff x \in A$ and $x \notin B$

$\iff x \in B$ and $x \notin B$ (∵ by (4))

$\iff x \in B$ and $x \in B^c$

$\iff x \in (B \cap B^c)$

$\iff x \notin \phi$

$\therefore \boxed{A - B = \phi}$ H.P.

(c) To show $A \cup B = A \cap B \iff A = B$

Consider $A = B$ ——(5)

we show that $A \cup B = A \cap B$

let x be the any arbitrary element such that

$x \in A \cup B \iff x \in A$ OR $x \in B$

$\iff x \in A$ OR $x \in A$ (∵ By (5))

$\iff x \in A$

$\iff x \in A$ and $x \in A$

$\iff x \in A$ and $x \in B$ (∵ By (5))

$\iff x \in A \cap B$

$\therefore \boxed{A \cup B = A \cap B}$

Conversely! let us consider $A \cup B = A \cap B$ ——(6)

we show that $A = B$

let x be the any arbitrary element from the universal set X. and A & B are subset of X.

Thus either $x \in A$ OR $x \in B$

$\Rightarrow x \in A \cup B$

$\Rightarrow x \in A \cap B$ (∵ given (6))

Scanned by CamScanner

$\Rightarrow x \in A$ and $x \in B$

$\Rightarrow x \in B$

$\therefore$ If $A \cup B = A \cap B$, then for every $x \in A, \Rightarrow x \in B$

$\therefore \boxed{A = B}$ H.P.

✱ Symmetric Difference of sets :⇒

Let A and B are any two non empty sets, then their symmetric difference is denoted and define as

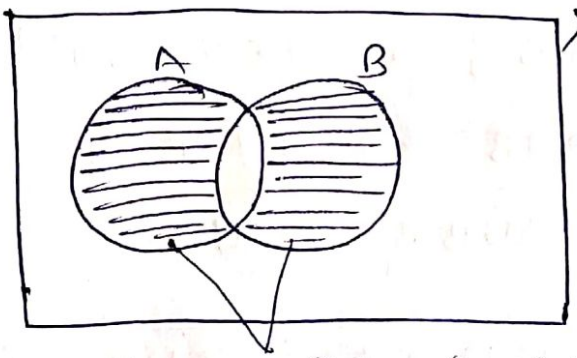
$$A \oplus B = \{ x \mid x \in A \cup B \text{ and } x \notin A \cap B \}$$

i.e. $\boxed{A \oplus B = (A \cup B) - (A \cap B)}$

i.e. $\boxed{A \oplus B = (A - B) \cup (B - A)}$

Thus $A \oplus B$ (the symmetric difference of set A & B) is the set of elements which either belongs to set A or set B but not belongs to both the sets.. (ie not belong to $A \cap B$)

Venn Diagram



$A \oplus B = (A - B) \cup (B - A)$

$= (A \cup B) - (A \cap B)$

Scanned by CamScanner

Example (11) Let A and B are two non-empty sets Then prove that

$$(A-B) \cup (B-A) = (A \cup B) - (A \cap B) = A \oplus B$$

Soln: Let x be the any arbitrary element such that

$x \in (A-B) \cup (B-A)$

$\Leftrightarrow [x \in A-B]$ OR $[x \in B-A]$

$\Leftrightarrow [x \in A \text{ and } x \notin B]$ OR $[x \in B \text{ and } x \notin A]$

$\Leftrightarrow [x \in A \text{ and } x \in B^c]$ OR $[x \in B \text{ and } x \in A^c]$

$\Leftrightarrow [(x \in A \text{ and } x \in B^c) \text{ OR } x \in B]$ and $[[(x \in A \text{ and } x \in B^c) \text{ OR } x \in A^c]$

$\Leftrightarrow [(x \in A \underline{\text{OR}} x \in B) \text{ and } (x \in B^c \text{ OR } x \in B)]$

and $[(x \in A \text{ OR } x \in A^c) \text{ and } (x \in B^c \underline{\text{OR}} x \in A^c)]$

$\Leftrightarrow [x \in A \cup B \text{ and } x \in B^c \cup B]$ and $[x \in A \cup A^c \text{ and } x \in B^c \cup A^c]$

$\Leftrightarrow [x \in A \cup B \text{ and } x \in B^c \cup B]$ and $[x \in A \cup A^c \text{ and } x \in (B \cap A)^c]$

$\Leftrightarrow [x \in A \cup B \text{ and } x \in X]$ and $[x \in X \text{ and } x \in (A \cap B)^c]$ [∵ By Demorgan's law]

$\Leftrightarrow [x \in (A \cup B) \cap X]$ and $[x \in (A \cap B)^c \cap X]$ [∵ X is universal set]

$\Leftrightarrow [x \in A \cup B]$ and $[x \in (A \cap B)^c]$

[For any $S \subseteq X$ ∵ $X \cap S = A$]

$\Leftrightarrow x \in A \cup B$ and $x \notin A \cap B$

$\Leftrightarrow x \in (A \cup B) - (A \cap B)$.

$\therefore \boxed{(A-B) \cup (B-A) = (A \cup B) - (A \cap B)}$ H.P.

Scanned by CamScanner

Example (12):

Prove that $A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)$

Sol^n: $\Rightarrow$ let x be the arbitrary element such that

$x \in A \cap (B \oplus C)$

$\Leftrightarrow x \in A$ and $x \in (B \oplus C)$

$\Leftrightarrow x \in A$ and $x \in [(B \cup C) - (B \cap C)]$

$\Leftrightarrow x \in A$ and $[x \in (B \cup C)$ and $x \notin (B \cap C)]$

$\Leftrightarrow [x \in A$ and $x \in (B \cup C)]$ and $[x \in A$ and $x \notin (B \cap C)]$

$\Leftrightarrow [x \in A$ and $(x \in B$ OR $x \in C)]$ and $[x \in A$ and $x \in (B \cap C)^c]$

$\Leftrightarrow [x \in A$ and $(x \in B$ OR $x \in C)]$ and $[x \in A$ and $x \in (B^c \cup C^c)]$

$\Leftrightarrow \{[x \in A$ and $x \in B]$ OR $[x \in A$ and $x \in C]\}$

and $\{[x \in A$ and $x \in B^c]$ OR $[x \in A$ and $x \in C^c]\}$

$\Leftrightarrow [(x \in A \cap B)$ OR $(x \in A \cap C)]$ and $[(x \in (A \cap B^c)$ OR $(x \in A \cap$

$\Leftrightarrow [x \in (A \cap B) \cup (A \cap C)]$ and $\{[x \in A$ and $x \in B^c]$

OR $[x \in A$ and $x \in C^c]\}$

$\Leftrightarrow [x \in (A \cap B) \cup (A \cap C)]$ and $\{[x \in A$ and $x \notin B]$ OR $[x \in A$ and $x \notin$

$\Leftrightarrow [x \in (A \cap B) \cup (A \cap C)]$ and $[x \notin A \cap B$ OR $x \notin A \cap C]$ #

$\Leftrightarrow [x \in (A \cap B) \cup (A \cap C)]$ and $[x \notin (A \cap B) \cap (A \cap C)]$

$\Leftrightarrow x \in [(A \cap B) \cup (A \cap C)] - [(A \cap B) \cap (A \cap C)]$

$\Leftrightarrow x \in (A \cap B) \oplus (A \cap C)$ $\quad (\because (M \cup N) - (M \cap N) = M \oplus N)$

$\therefore \boxed{A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)}$ Hp.

Scanned by CamScanner

H.W:

① Prove the followings

(a) $A \cap (B-A) = \phi$

(b) $(A-B) \cap B = \phi$

(c) $A \cup B = (A-B) \cup B$

(d) $A-B = A \cap B^c$

(e) $A-B = B^c - A^c$

(f) $A-(A-B) = A \cap B$

(g) $(A-C) \cap (B-C) = (A \cap B)-C$

(h) $A \cap (B-C) = (A \cap B)-(A \cap C)$

(i) $A \cup (B-C) \neq (A \cup B)-(A \cup C)$

Scanned by CamScanner

★ Mathematical Logic ★

• Statement OR proposition :⇒

A statement OR proposition is a declarative sentence that is either true or false but not both. and the value associated with truthfulness o r falsity of a statement is called as its truth value.

For Convenience, an english alphabet (lower case) p, q, r, s.... etc are use to denote propositional variables that can be replace by statement.

for eg: ① p : It is raining

② q : It is cold

③ s : $\{2,3\} \subseteq \{1,2,3,4\}$

④ r : $3+3 = 7$ etc.

• Compound statement :⇒

Two or more simple propositions can be combined by means of logical connectives (operator) to form a single proposition called as compound statement.

There are five logical connectives which are Foftenly used for this purpose.

Connectives	symbol	Name
not	$\sim$ ($\neg$)	Negation
and	$\wedge$	Conjuction
or	$\vee$	Disjunction
If — then	$\rightarrow$	Implication (Conclitional)
If and only If	$\leftrightarrow$	equivalance (Bi Conditional)

Scanned by CamScanner

Basic Logical operations

(1) Negation

If p is a statement then negation of p is denoted by $\sim p$ (OR $\neg p$) and it is read as "not p"

Truth table

p	~p
T	F
F	T

For eg: p : Delhi is capital of India

~ p : Delhi is not capital of India.

(2) Conjuction :

When two (or more) statements are joined by the connectives "and" then compound statement is called ~~Compound~~ conjunction of ~~that~~ these sub-statements. Symbically it is denoted by "$\wedge$"

ie. Let p & q are any two statements then the statement "p" and "q" is called conjuction of p & q and denoted as "$p \wedge q$"

Truth Table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Scanned by CamScanner

For eg: let p: Delhi is capital of India
q: 8+5 = 13

Then $p \wedge q$: Delhi is capital of India and 8+5 = 13.

If p and q both have truth values as T. then '$p \wedge q$' is also have truth value T.

But if at least one statement ~~of~~ of out of two statements p and q are false then compound statement '$p \wedge q$" is always false.

For ex: p: Delhi is capital of India (T)
q: 8+5 ≠ 13 (F)

Then $p \wedge q$: Delhi is capital of India and 8+5 ≠ 13 is false (F)

---

(3) Disjunction:

When two (or more) statements are joined by the connectives 'OR' then compound statement is called as disjuction of these sub statements. Symbolically it is denoted by 'V'

ie. let p and q are any two statements then the statement "p or q" is called disjuction of p & q.

and it is denoted as '$p \vee q$'

Truth table.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Scanned by CamScanner

For eg: P: 5 is positive integer

Q: 6−1 = 5

Then P ∨ Q : 5 is positive integer OR 6−1 = 5

If P and Q both that truth value F then P∨Q is also False(F). But ~~any one of the~~ If at least one of the statements (i.e. P and Q) are having truth value F then P∨Q is always truth(T)

For eg: P : 5 is odd number (T)

Q : 6−1 ≠ 5 (F)

Then P ∨ Q : 5 is odd number OR 6−1 ≠ 5 (T)

---

(4) Conditional (Implication)

If two statements p and q connected with logical connectives 'If − then" then the compound statement is known as conditional statement and symbolically it is denoted as "$p \rightarrow q$" and it is read as If "P" then "Q".

For eg: Here statement P is called as Hypothesis and Q is called as conclusion

A conditional statement is used as a guarantee for the conclusion to be true provided hypothesis is true

ie (i) if P (Hypothesis) is True and Q (conclusion) is true then $p \rightarrow q$ is true

(ii) If P (Hypothesis) is true but Q (conclusion) is false then $p \rightarrow q$ is false

Scanned by CamScanner

(iii) If p (Hypothesis) is false (F) and ~~conclusi~~ Conclusion q is either T or F then $p \to q$ is always true

Truth table

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

For eg: (i) Let p: $3+9 = 12$ (T)

q: $12-9 = 3$ (T)

Here $p \to q$: If $3+9=12$ then $12-9=3$ (T)

(ii) Let p: $2+4 = 6$ (T)

q: $3+3 = 0$ (F)

Here $p \to q$: If $2+4=6$ then $3+3=0$ (F)

(iii) let p: $2+4 = 10$ (F)

q: $3+2 = 5$ (T)

Here $p \to q$: If $2+4=10$ then $3+2=5$ (T)

(iv) let p: $3-1 = 8$ (F)

q: $4+1 = 7$ (F)

Here $p \to q$: If $3-1=8$ then $4+1=7$ (T)

Scanned by CamScanner

(IV) Biconditional (Equivalence)

Let p and q are two statements connected with logical connectives "if and only if" then the compound statement is known as Biconditional or equivalence statement. Symbolically it is denoted as "$p \leftrightarrow q$" and read as p if and only if q

A Biconditional statement is always true if both of the statements p and q are either true or false otherwise Biconditional statement is always false.

Truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

For eg: let p : $5+2=7$

q : $7-2=5$

Then $p \leftrightarrow q$: $5+2=7$ if and only if $7-2=5$

Scanned by CamScanner

Example: Construct a truth table for the following statements

(i) $P \wedge (\sim q \vee q)$

(ii) $(P \vee q) \rightarrow \sim(P \wedge \sim q)$

Soln (i) $P \wedge (\sim q \vee q)$

1	2	3	4 : (2 ∨ 3)	5 : (1 ∧ 4)
P	q	~q	(~q ∨ q)	P ∧ (~q ∨ q)
T	T	F	T	T
T	F	T	T	T
F	T	F	T	F
F	F	T	T	F

(ii) $(P \vee q) \rightarrow \sim(P \wedge \sim q)$

P	q	P ∨ q	~q	(P ∧ ~q)	~(P ∧ ~q)	(P ∨ q) → ~(P ∧ ~q)
T	T	T	F	F	T	T
T	F	T	T	T	F	F
F	T	T	F	F	T	T
F	F	F	T	F	T	T

Scanned by CamScanner

Derived connectives

(1) NAND [not And]

It is the negation of compound statement $(P \wedge q)$ and it is denoted as $(P \uparrow q)$

ie $P \uparrow q \equiv N(P \wedge q)$

(2) NOR [not OR]

It is the negation of compound statement "$P \vee q$" and it is denoted as $P \downarrow q$

ie. $P \downarrow q \equiv \sim(P \vee q)$

(3) XOR (not. If and only If)

It is the negation of compound statement "$P \leftrightarrow q$" and it is denoted as $P \oplus q$

ie. $P \oplus q \equiv \sim(P \leftrightarrow q)$

Scanned by CamScanner

Logical Equivalance :

The propositions $S_1(p, q, r, \ldots)$ and $S_2(p, q, r, \ldots)$ are said to be logically equivalant or logically same if both S_1 and S_2 have identical truth table and it is denoted as $S_1 \equiv S_2$

where, $p, q, r \ldots$ etc are propositional variables & S_1 & S_2 are compound statements.

Foreg: Show that

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

Soln: let $S_1 = \sim(p \wedge q)$

& $S_2 = \sim p \vee \sim q$

Truth table

(1)	(2)	(3)	(4)	(5)	(6)	(7)
p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \vee \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

From the column (6) & (7), since the truth value of $S_1 = \sim(p \wedge q)$ & $S_2 : \sim p \vee \sim q$ are identical

$\therefore$ S_1 is equivalant with S_2

ie $\boxed{\sim(p \wedge q) \equiv \sim p \vee \sim q}$

Scanned by CamScanner

Ex: Show that

$$P \vee (q \wedge r) \equiv (P \vee q) \wedge (P \vee r)$$

Soln: Let p, q, r are propositional variables

and $S_1 = P \vee (q \wedge r)$

& $S_2 = (P \vee q) \wedge (P \vee r)$ are two compound statements.

By using truth table

①	②	③	④	⑤	⑥	⑦	⑧
p	q	r	$P \vee q$	$P \vee r$	$q \wedge r$	$P \vee (q \wedge r)$	$(P \vee q) \wedge (P \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	T	F	T	T
T	F	F	T	T	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	F	F
F	F	T	F	T	F	F	F
F	F	F	F	F	F	F	F

From column ⑦ & ⑧, since the truth table of statement $S_1 = P \vee (q \wedge r)$ & $S_2 = (P \vee q) \wedge (P \vee r)$ having identical truth values.

$$\therefore \quad S_1 \equiv S_2$$

i.e $\boxed{P \vee (q \wedge r) \equiv (P \vee q) \wedge (P \vee r)}$

Scanned by CamScanner

Ex: Show that the following statements are logically equivalent.

(1) $p \vee (q \vee r) \equiv (p \vee q) \vee r$

(2) $p \vee \sim q \equiv (p \vee q) \wedge \sim(p \wedge q)$

Note :

(1) Negation of Negation

$$\sim(\sim p) \equiv p$$

(2) Negation of conjuction $(p \wedge q)$

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

(3) Negation of Disjunction $(p \vee q)$

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

(4) Negation of Implication $(p \to q)$

$$\sim(p \to q) \equiv p \wedge \sim q$$

(5) Negation of Biconditial $(p \leftrightarrow q)$

$$\sim(p \leftrightarrow q) \equiv p \leftrightarrow \sim q$$
$$\equiv \sim p \leftrightarrow q$$

Scanned by CamScanner

Converse, Inverse and Contrapositive of Statements

If "$p \rightarrow q$" is any conditional statement then its

(i)	inverse is $(\sim p \rightarrow \sim q)$
(ii)	converse is $(q \rightarrow p)$
(iii)	Contrapositive is $(\sim q \rightarrow \sim p)$

★ Tautology :

A compound statement $S(p, q, r \ldots)$ that is always true for all truth values of its variables p, q, r etc is called as Tautology.

In other word, If in the truth table of statement $S(p, q, r \ldots)$ all values are only T. then Statement $S(p, q, r \ldots)$ is Tautology.

For eg: let $S : p \vee \sim p$

p	$\sim p$	$p \vee \sim p$
T	F	T
T	F	T

$\therefore$ $p \vee \sim p$ is Tautology.

Scanned by CamScanner

✱ Contradiction (Absurdity)

If a compound statement $S(P, Q, R \ldots)$ is always false (F) for any truth value of its variables then S is called as contradiction. or absurdity.

For eg: $S : P \wedge (Q \wedge \sim P)$

P	Q	$(\sim P)$	$(Q \wedge \sim P)$	$P \wedge (Q \wedge \sim P)$
T	T	F	F	F
T	F	F	F	F
F	T	T	F	F
F	F	T	F	F

$\therefore P \wedge (Q \wedge \sim P)$ is contradiction.

✱ ~~Conting~~ Contingency :

A compound statement which is neither tautology nor contradiction is known as contingency.

For eg : $S = P \wedge Q$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

$\therefore P \wedge Q$ is Contingency.

Scanned by CamScanner

Example: Write converse, Inverse, contrapositive and a negation of following statements

(i) If it is raining, the grass is wet

(ii) Rain is sufficient for it to be cloudy.

(iii) "Oxygen in the air decreases, as we go up"

Soln: (i) Given statement is

"If It is raining, the grass is wet"

let p : It is raining

q : The grass is wet

∴ given statement is symbolically written as

$p \to q$

∴ Converse of $p \to q$ is "$q \to p$"

ie. "If grass is wet then it is raining"

Inverse of $p \to q$ is $\sim p \to \sim q$

"If it is not raining then the grass is not wet"

Contrapositive of $p \to q$ is $\sim q \to \sim p$

"If the grass is not wet then it is not raining

Negation of $p \to q$ is $p \wedge \sim q$

"It is raining and the grass is not wet"

Scanned by CamScanner

(II) "Rain is sufficient for it to be cloudy."

let p : It is raining
q : It is cloudy

given statement can be rewrite as

"If it is raining then it is cloudy"

ie $p \rightarrow q$.

∴ Converse $(q \rightarrow p)$

"If it is cloudy then it is raining"

Inverse $(\sim p \rightarrow \sim q)$

"If it is not ~~cloudy~~ raining then it is not ~~raining~~ cloudy"

Contrapositive $(\sim q \rightarrow \sim p)$

"If it is not cloudy then it is not raining"

Negation $(p \wedge \sim q)$

"It is raining and it is not cloudy."

(III) "Oxygen in the air decreases as we go up"

let p : we go up
q : oxygen in air decreases.

the above statement can be rewrite as

"If we go up then oxygen in the air decreases"

ie $p \rightarrow q$

Scanned by CamScanner

∴ Converse $(q \to p)$

"If oxygen in the air decrease then we go up"

Inverse $(\sim p \to \sim q)$

"If we do not go up then oxygen in the air not decreses."

Contrapositive. $(\sim q \to \sim p)$

"If oxygen in the air not decreases then we do not go up"

Negation: $(p \wedge \sim q)$

we go up and oxygen in the adr not decreases.

Scanned by CamScanner

Arguments :⇒

An argument is an assertion that a given set of propositions $P_1, P_2, P_3 \ldots P_n$ (ie premises) gives a consequence ie. conclusion Q. [another proposition]

Such argument is denoted by

$$P_1, P_2, \ldots P_n \vdash Q$$

- An argument $P_1, P_2, \ldots P_n \vdash Q$ is valid if and only if a statement

  $S : (P_1 \wedge P_2 \wedge \ldots \wedge P_n \Rightarrow Q)$ is Tautology

- An argument which is not valid is called as Fallacy.

Ex: Test the validity of the following ~~stat~~ argument.

"If I study, I will not fail in DMPT

If I do not play cricket, I will study

But I fail in DMPT.

Therefore, I must have played cricket.

Soln: Given propositional variables (premises) are

p : I study

q : I will fail in DMPT

r : I play cricket.

Scanned by CamScanner

-Thus given arguments are

$P_1 : P \to \sim q$

$P_2 : \sim r \to p$

$P_3 : q$

$\therefore Q : r$

To prove given argument is valid we need to show that

$P_1 \wedge P_2 \wedge P_3 \Longrightarrow Q$ is Tautology.

ie. $[(P \to \sim q) \wedge (\sim r \to p) \wedge q] \Rightarrow r$ is Tautology.

Truth table.

①	②	③	④	⑤	⑥	⑦	⑧	⑨	10
p	q	r	$\sim q$	$\sim r$	$p \to \sim q$	$\sim r \to p$	⑥∧⑦	⑧∧②	⑨→③
T	T	T	F	F	F	T	F	F	T
T	T	F	F	T	F	T	F	F	T
T	F	T	T	F	T	T	T	F	T
T	F	F	T	T	T	T	T	F	T
F	T	T	F	F	T	T	T	T	T
F	T	F	F	T	T	F	F	F	T
F	F	T	T	F	T	T	T	F	T
F	F	F	T	T	T	F	F	F	T

Since all the truth values in column ⑩

ie. $(P \wedge \sim q) \wedge (\sim r \to p) \wedge q \Longrightarrow r$ are true

∴ given argument is tautology. Hence it is valid.

Scanned by CamScanner

Example: Test the validity of the following argument

All my friends are musicians

John is my friend

None of my neighbors are musicians

∴ John is not my neighbor.

Soln: Let propositional variables are

P: they are my friends

Q: They are musician.

R: they are neighbors

∴ The given argument is written as

$P_1 : P \wedge Q$

$P_2 : P$

$P_3 : R \wedge \sim Q$

$\therefore Q : \sim R$

To prove the given argument is valid.

ie $P_1, P_2, P_3 \vdash Q$ is valid

we need to show that

$(P_1 \wedge P_2 \wedge P_3) \longrightarrow Q$ is Tautology

ie. $[(P \wedge Q) \wedge P \wedge (R \wedge \sim Q)] \longrightarrow \sim R$ is Tautology.

Scanned by CamScanner

Truth table

①	②	③	④	⑤	⑥	⑦	⑧ = ⑥ AND	⑨	10
P	Q	R	$\sim Q$	$\sim R$	$P \wedge Q$	$R \wedge \sim Q$	$(P \wedge Q) \wedge P$	⑧ ∧ ⑦	⑨ → ⑤
T	T	T	F	F	T	F	T	F	T
T	T	F	F	T	T	F	T	F	T
T	F	T	T	F	F	F	F	F	T
T	F	F	T	T	F	F	F	F	T
F	T	T	F	F	F	F	F	F	T
F	T	F	F	T	F	F	F	F	T
F	F	T	T	F	F	F	F	F	T
F	F	F	T	T	F	F	F	F	T

Since all the truth value in column ⑩ are T.

i.e. argument $[(P \wedge Q) \wedge P \wedge (R \wedge \sim Q)] \rightarrow \sim R$

is tautology.

(H.W) Test the validity of following arguments

① "Either Ram will cook or Hari will practice karate"
"If Hari practice karate then Ram studies"
Ram do not study
Therefore, Hari will practice karate (Invalid)

---

② If a man is a bachelor, he is unhappy
If a man is unhappy, he dies young.
∴ Bachelors dies yound. (valid)

---

③ If 7 is less than 4 then 7 is not prime number
7 is not less than 4
∴ 7 is a prime number (Invalid)

Scanned by CamScanner

Ex: ~~Test weather for the~~

Test wheather the following statements are tutology, contradiction or ~~contingency~~.

① $[\sim q \wedge (p \to q)] \to \sim p$ (Tutology)

② $[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$ (Tutology)

③ $(p \wedge q) \wedge \sim (p \vee q)$ (contradiction)

Scanned by CamScanner

✱ Principle of Mathematical Induction ∴ ⇒

Let $P(n)$ be a statement or proposition defined by the set of positive integers. Such that it is either true or flase for all $n \in I^+$

To prove given statement $P(n)$ is true or valid we need to prove that

(i) $P(n)$ is true for $n=1$ [i.e. $P(1)$ is true]

and then by assuming $P(k)$ for $k \geq 1$ is true

we need to prove that

(ii) $P(k+1)$ for $n=k+1$ is also true

In this way we can conclude that $P(n)$ is true for all integers $n \in I^+$.

Here first proposition (i) is usually refferred as the basis of Induction. and the proposition (ii) ie. $P(k+1)$ is usually refferred as Induction step.

Ex: Using Mathematical Induction method, prove that

$$1+3+6+\cdots+\frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

for all $n \in I^+$

Soln: let us consider

$$P(n): 1+3+6+\cdots+\frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

be the given statement.

Scanned by CamScanner

for $n=1$.

$$P(1) = 1 \quad \& \quad RHS = \frac{1(1+1)(1+2)}{6} = \frac{6}{6} = 1$$

$\therefore$ $P(n)$ is true for $n=1$

Now To prove for any ~~$k \geq 1$ and~~ $n = k+1$ (where $k \geq 1$)

$P(k+1)$ is also true

we assume that $P(k)$ is true

ie $P(k)$: $1 + 3 + 6 + \cdots + \frac{k(k+1)}{2} = \frac{k(k+1)(k+2)}{6}$ ——— (1)

Now.

$$P(k+1) = 1 + 3 + 6 + \cdots + \frac{(k+1)[(k+1)+1]}{2}$$

$$= \left[1 + 3 + 6 + \cdots + \frac{k[k+1]}{2}\right] + \frac{(k+1)[(k+1)+1]}{2}$$

$$= P(k) + \frac{(k+1)[(k+1)+1]}{2}$$

$$= \frac{k(k+1)(k+2)}{6} + \frac{(k+1)[k+2]}{2} \quad (\because \text{from (1)})$$

$$= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{6}$$

$$= \frac{(k+1)(k+2)(k+3)}{6}$$

$$= \frac{(k+1)[(k+1)+1][(k+1)+2]}{6}$$

$\Rightarrow$ $P(k+1)$ is also true

Thus $P(n)$ is true for all $n \in \mathbb{N}$

ie $$\boxed{1 + 3 + 6 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}}$$ Hip.

Scanned by CamScanner

Example: Using Principle of Mathematical induction prove that

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$$

Soln: let $P(n)=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n(n+1)}$ (where $n\in I^+$)

For $n=1$:

$P(1)$: $\frac{1}{1\cdot 2}=1/2$

& R.H.S $=\frac{1}{1+1}=1/2$

$\Rightarrow$ $P(1)$ is true

ie $P(n)$ is true for $n=1$.

To prove that $P(k+1)$ is true for $n=k+1$, (where $k\geq 1$)

we assume that $P(k)$ is true (For $n=k$)

ie $$P(k)=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}=\frac{k}{k+1} \quad \text{—(1)}$$

Now.

$$P(k+1)=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k+1)[(k+1)+1]}$$

$$=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}+\frac{1}{(k+1)[(k+1)+1]}$$

$$=P(k)+\frac{1}{(k+1)[(k+1)+1]} \quad \because \text{(from (1))}$$

$$=\frac{k}{(k+1)}+\frac{1}{(k+1)[(k+1)+1]}$$

$$=\left[\frac{k}{(k+1)}+\frac{1}{(k+1)(k+2)}\right]=\frac{k(k+2)+1}{(k+1)(k+2)}$$

$$=\frac{k^2+2k+1}{(k+1)(k+2)}=\frac{(k+1)(k+1)}{(k+1)(k+2)}=\frac{k+1}{(k+2)}$$

$$P(k+1)=\frac{(k+1)}{(k+1)+1}$$

$\therefore$ $P(n)$ is true for $n=k+1$ where $k\geq 1$

$\therefore$ $P(n)$ is true for all $n\in I^+$.

ie $$\boxed{\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}}$$ H.P.

Scanned by CamScanner

Example: Using Mathematical induction method show that

$$1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

Soln: let $P(n) : 1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$

For $n=1$: $P(1) = 1^2$ RHS $= \frac{1(1+1)(2(1)+1)}{6} = 1$

$\Rightarrow$ $P(n)$ is true for $n=1$.

To show $P(k+1)$ For $n=k+1$ (where $k \geq 1$) is also true

we assume that $P(n)$ is true for $n=k$

ie. $P(k) = 1^2+2^2+3^2+\cdots+k^2 = \frac{k(k+1)(2k+1)}{6}$ —①

Now

$P(k+1)$: $1^2+2^2+3^2+\cdots+(k+1)^2$

$= \left(1^2+2^2+3^2+\cdots+k^2\right) + (k+1)^2$

$= P(k) + (k+1)^2$ (from ①)

$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$\Rightarrow \frac{(k+1)\left[k(2k+1)+6(k+1)\right]}{6}$

$= \frac{(k+1)\left[2k^2+k+6k+6\right]}{6}$

$= \frac{(k+1)\left[2k^2+7k+6\right]}{6} = \frac{(k+1)(k+2)(2k+3)}{6}$

$= \frac{(k+1)\left[(k+1)+1\right]\cdot\left[2(k+1)+1\right]}{6}$

$\therefore P(n)$ is true for $n=k+1$.

$\therefore \Rightarrow P(n)$ is true for all $n \in I^+$

ie. $\boxed{1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}}$ H.P

Scanned by CamScanner

Example: By method of mathematical induction

Show that

$$1 + x + x^2 + \cdots\cdots + x^{n-1} = \frac{x^n - 1}{x-1}, \text{ for } x > 1$$

Soln: let us consider

$$P(n) = 1 + x + x^2 + \cdots\cdots + x^{n-1} = \frac{x^n - 1}{x-1}$$

& RHS $= \frac{x^n - 1}{x-1}$,

for: $n=1$: $P(1) = 1$ & RHS $= \frac{x^1 - 1}{x-1} = \frac{x-1}{x-1} = 1$

$\therefore$ $P(n)$ is true for $n=1$.

To show $P(n)$ is true for $n=k+1$, (where $k \geq 1$)

we assume that

$$P(k) : 1 + x + x^2 + \cdots + x^{k-1} = \frac{x^k - 1}{x-1} \quad \text{is true} \quad \text{——— (1)}$$

Now. For $n = k+1$

$$P(k+1) = 1 + x + x^2 + \cdots\cdots + x^{(k+1)-1}$$

$$= \left(1 + x + x^2 + x^3 + \cdots\cdots + x^{k-1}\right) + x^k$$

$$= P(k) + x^k$$

$$= \left(\frac{x^k - 1}{x-1}\right) + x^k \qquad (\because \text{from (1)})$$

$$= \frac{(x^k - 1) + x^k(x-1)}{(x-1)} = \frac{(x^k - 1 + x^{k+1} - x^k)}{(x-1)}$$

$$= \frac{x^{k+1} - 1}{x-1} = \text{RHS}$$

$\Rightarrow$ $P(n)$ is true for $n = k+1$

$\therefore$ $P(n)$ is true for all integer $n \in I^+$.

ie $\boxed{1 + x + x^2 + \cdots\cdots + x^{n-1} = \frac{x^n - 1}{x-1}}$ H.P.

Scanned by CamScanner

Ex: Using mathematical induction method, prove that

$$n < 2^n \quad \forall\, n \in I^+$$

Soln: Let us consider

$$P(n) = n \quad \& \quad RHS = 2^n$$

for $n=1$: $P(1) = 1$ & $RHS = 2^1 = 2$

Since $1 < 2$ i.e. $P(1) < 2^1$

$\therefore$ $P(n)$ is true for $n=1$

To prove $P(k+1)$ is True (For $n=k+1$, where $k \geq 1$)

we ~~Show that~~ assume that $P(k)$ is true

i.e. $P(k)$: $k < 2^k$ ——— (1)

Now,

$$P(k+1) = k+1$$

$$< 2^k + 1 \quad \because \text{from (1)}$$

$$< 2^k + 1 + 1 \quad \text{(It is obvious)} \because 1 < 2$$

$$< 2^k + 2$$

$$< 2^k \cdot 2^k \quad \text{for } k \geq 1$$

$$= 2^{k+1}$$

$\therefore$ $P(k+1)$: $k+1 < 2^{k+1}$

Thus $P(n)$ is true for $n = k+1$, where $k \geq 1$

$\therefore$ $P(n)$ is true for all $n \in I^+$

i.e. $\boxed{n < 2^n}$ Hence proved.

Scanned by CamScanner

Example: Using mathematical induction method, prove that

(a) If $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ then $A^n = \begin{bmatrix} 1 & na \\ 0 & 1 \end{bmatrix}$

Soln: let $P(n) : A^n$ & R.H.S. $= \begin{bmatrix} 1 & na \\ 0 & 1 \end{bmatrix}$

For $n=1$.

$P(1) = A^1 = A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ (given)

& RHS $= \begin{bmatrix} 1 & 1a \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$

$\therefore \Rightarrow P(n)$ is true for $n=1$

To show $P(n)$ is true for $n=k+1$, (where $k \geq 1$)

We assume that $P(k)$ [for $n=k$] is true

ie $P(k) : A^k = \begin{bmatrix} 1 & ka \\ 0 & 1 \end{bmatrix}$ ——— (1)

Now

$P(k+1) = A^{k+1} = A^k \cdot A$

$= \begin{bmatrix} 1 & ka \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ (∵ from (1))

$= \begin{bmatrix} 1+0 & a+ka \\ 0+0 & 0+1 \end{bmatrix}$

$= \begin{bmatrix} 1 & (k+1)a \\ 0 & 1 \end{bmatrix}$

$\Rightarrow P(n)$ is also true for $(n=k+1)$ where $k \geq 1$

$\therefore P(n)$ is true for all $n \in \mathbb{I}^+$

ie If $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ then $A^n = \begin{bmatrix} 1 & na \\ 0 & 1 \end{bmatrix}$ is Holds.

Scanned by CamScanner

Ex:1 Using mathematical induction method show that for $n \geq 1$, n^3+2n is divisible by 3

Soln: Let

$$P(n) : n^3+2n$$

For $n=1$: $P(1) : 1^3+2(1) = 1+2 = 3$

and it is divisible by 3.

$\therefore P(n)$ is true for $n=1$

Now, To show $P(k+1)$ is true (for $n=k+1$, where $k \geq 1$)

we assume that $P(k)$ is true

i.e (k^3+2k) is divisible by 3. ——— (1)

Now

$$P(k+1) : (k+1)^3+2(k+1)$$

$$\Rightarrow k^3+3k^2+3k+1+2k+2$$

$$= k^3+3k^2+3k+2k+3$$

$$= (k^2+2k)+(3k^2+3k+3)$$

$$= (k^2+2k)+3(k^2+k+1)$$

Since $k \geq 1$ is any integer, and thus k^2+k+1 is any const.

$\therefore 3(k^2+k+1)$ is divisible by 3

also k^2+2k is divisible by 3 (By-(1))

$\therefore P(k+1) : (k^2+2k)+3(k^2+k+1)$ is divisible by 3

$\Rightarrow P(n)$ is true for $n=k+1$, wher $k \geq 1$

$\therefore$ **$P(n) : n^3+2n$, $\forall n \geq 1$ is divisible by 3** is true

Scanned by CamScanner

H.W

Using mathematical induction Method prove the followings

① $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} = \left(1 - \frac{1}{2^n}\right)$

② $\frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$

③ $a + ar + ar^2 + \cdots + ar^{n-1} = a\left(\frac{r^n - 1}{r-1}\right)$ if $r \neq 1$

④ $1\cdot2 + 2\cdot3 + 3\cdot4 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$

⑤ $1\cdot3 + 3\cdot5 + 5\cdot7 + \cdots + (2n-1)(2n+1) = \frac{n(4n^2+6n-1)}{3}$

⑥ $1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$

Scanned by CamScanner